Machine Fault Classification Using Random Forest and XGBoost on NASA CMAPSS Jet Engine Dataset

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***Abstract*—Predictive maintenance is one of the excellent research topic in industrial systems because that an unexpected machine failure would lead a costly maintenance, wasted a lots of time and energy, operational downtime, security problems and hazards. In this paper we show a machine learning based method for machine fault classification using NASA CMAPSS turbofan engine degradation dataset and the dataset itself was provided by NASA. The main objective of our work is to identify and predict the faults by classifying the condition of the engine to the 'normal' and 'faulty' based on measurements from sensors and the operational data. Our proposed methodology comprises the following steps; data preprocessing, RUL prediction, fault labeling, Exploratory data analysis, class balancing by using SMOTE, and finally two ensemble learning models training; Random Forest and XGBoost, by using Hyperparameter tuning, and classification metrics based evaluation like accuracy, precision, recall, and F1-score . The experimental results indicate that the two models predict with high classification performance, and XGBoost offers a slightly better capability of generalization and classification. The major contributions of this work are the proposed full predictive maintenance pipeline, the comparative evaluation of ensemble learning algorithms for fault detection, and the study on feature importance of sensors on machine degradation. It can be verified that machine learning methods have effectively shown the capability of smart fault diagnosis and predictive maintenance.**

***Keywords*—Machine Fault Classification, NASA CMAPSS Dataset, Random Forest, XGBoost, Machine Learning, Fault Diagnosis, Remaining Useful Life (RUL), SMOTE.**

I. Introduction

During our time industrial systems have become reliant on smart monitoring and maintenance strategy to maintain their operational reliability, reduce maintenance costs, time wastage and avoid the failures that are predicted to be fatal to the machinery. Where traditional systems have used a reactive/corrective system or an interval based/preventive system which either carry out maintenance only after a failure occurs or at pre-scheduled times regardless of the actual condition of the machinery. Time has moved forward industrial system have become more complex, more data driven and so a predictive maintenance system is the way to achieve optimum system availability.

As Machine learning and data learning techniques have developed so has the accuracy and efficiency of fault detection and diagnosis systems. Using Machine learning models the system is able to learn how normal operation looks and thus predict faults in system machinery by analyzing the various sensor readings and other operational data obtained from the equipment, for this research project two prominent supervised learning algorithms that we have chosen are Random Forest and XGBoost. Both are very predictive with accurate outputs and have a good capability to handle non-linear relationships [1].

The data we use in this research is NASA Commercial Modular Aero-Propulsion System Simulation (CMAPSS) dataset, which is widely considered as one of the benchmark datasets for predictive maintenance researches. This dataset provides multivariate time-series sensor readings from turbofan engines with varying degradations and fault types [2]. However, our used data set lacks the direct fault classification labels. It implies supervised learning algorithms cannot directly solve this problem. Therefore, Remaining Useful Life (RUL) estimation and fault label generation are both required as the prerequisite preprocess step.

In this research we propose to use Random Forest and XGBoost to develop machine learning models to classify faults using the NASA CMAPSS dataset. We involve various stages including: data preprocessing, RUL estimation, fault label generation, over-sampling the minority class using SMOTE technique, exploratory data analysis, hyperparameter tuning and model performance evaluation using several metrics. In addition, we show the comparison of two ensemble learning models on their efficiency in predictive maintenance tasks and demonstrate the significance of each sensor features concerning machine degradation.

The motivation of this research is to establish dependable intelligent maintenance systems which could provide early diagnosis on failures to support industrial decisions. By combining machine learning approaches and predictive analysis, the proposed framework could possibly improve reliability of the industrial equipment, lower the risks and strengthen maintenance strategy.

II. State of the Art

A. Related Work

This research article's title is ”Predictive Maintenance in Aircraft Engine Maintenance Using the CMAPSS Dataset: Performance Comparison and Evaluation of Machine Learning Classification Algorithms”, which discusses the use of machine learning algorithms for the purpose of the fault diagnosis in aircraft engines utilizing the CMAPSS data set. The predictive maintenance issue was converted to a binary classification problem, and several machine learning algorithms have been trained to predict faults and detect the occurrence of maintenance. The authors have tested many pre-processing techniques and performance evaluation criteria like accuracy, precision, recall, F1-score to assess how effective the developed algorithms were, and the finding proved that the machine learning algorithms could efficiently identify and detect degradation of the engine, facilitating the prediction based maintenance system in the industry [3].

Another paper with title ”Predictive maintenance on C-MAPSS using LSTM variants and attention mechanisms” proposed an approach using deep learning techniques in the context of predictive maintenance. The proposed system aims to conduct fault detection on a turbofan engine based on its degradation pattern by applying and studying multiple LSTM architectures augmented with attention mechanisms. The work uses a number of pre-processing and sequence learning techniques to capture temporal degradation from sensor data. The results showed that deep learning models are capable of learning non-linear degradation models and predict failures [4].

An investigation titled 'Explainable Data-Driven Method Coupled with Bayesian filtering for Aircraft Engine remaining useful life' applied a model for predictive maintenance using aircraft engine prognostics with the NASA CMAPSS dataset. It combined data-driven techniques, feature selection strategies, and Bayesian filtering methods to enhance remaining useful life prediction, as well as aircraft engine condition monitoring. Through various processing stages and optimization methods applied to the data to reduce the dimension of features, its prediction performance was increased. The experiment results showed the effectiveness of explainable ML in intelligent predictive maintenance and aircraft engine health monitoring system [5].

From the articles listed above it can be concluded that the NASA CMAPSS dataset is widely applied for aircraft engine fault diagnosis and predictive maintenance with a variety of ML and DL techniques. Early research papers were dedicated to fault detection, prediction of Remaining Useful Life, feature extraction and model tuning to obtain efficient maintenance decision and reliable engines. A number of algorithms ranging from classical machine learning classifiers, ensemble learning techniques to deep learning methods (e.g., LSTM networks) show promising results in terms of detecting the degradation patterns from sensor readings.

In light of the results in the literature review, this research endeavor aims to compare two ensemble learning algorithms; namely, Random Forest and XGBoost, for intelligent machine fault classification on NASA CMAPSS data. This work also entails data preprocessing, estimation of Remaining Useful Life, generation of fault labels, use of SMOTE to tackle class imbalance, and tuning of hyper-parameters to increase classification accuracy. An investigation will also be carried out to analyze how the two models perform in predictive maintenance scenarios and in early fault detection, as well as the impact of sensor data on engine degradation.

B. Presentation of the Algorithms

This work applies two ensemble machine learning techniques, namely Random Forest and XGBoost for intelligent machine fault classification of the NASA CMAPSS data. These methods were chosen as they exhibit excellent classification performance, are robust to noise and can model the complex nonlinear interdependencies between sensor data and machine degradation.

Random Forest is a supervised ensemble learning method which involves the construction of a set of decision trees during the training process. The individual trees are trained with randomly sampled instances from the data and randomly sampled features, and the prediction from the final Random Forest is determined by the majority vote from the set of all trees. This method effectively decreases overfitting and increases the generalization ability of a single decision tree. Random Forest is applied to various predictive maintenance applications because it is robust, it is a readily interpretable algorithm and it is possible to compute feature importance [6].

For the prediction function of the Random Forest we can expressed it in Equation (1):

$$\hat{y} = mode(h_1(x), h_2(x), ..., h_n(x)) \tag{1}$$

where:

- $\hat{y}$ represents the final predicted class
- $h_i(x)$ represents the prediction of the i^{th} decision tree
- x represents the input feature vector
- n represents the total number of decision trees
- $mode$ represents the majority voting operation

We also defined the Gini impurity which we used for node splitting in Equation (2):

$$Gini = 1 - \sum_{i=1}^{C} p_i^2 \tag{2}$$

where:

- $Gini$ represents the impurity measure
- C represents the number of classes
- p_i represents the probability of class i

XGBoost stands for Extreme Gradient Boosting. It's an advanced ensemble learning algorithm built upon gradient boosting machines. The Random Forest algorithm uses a collection of tree-like models that are independently built whereas. XGBoost algorithm trains trees sequentially so that the new tree would try to minimize prediction errors resulted by previous trees. Besides those traditional methods, XGBoost uses many other optimization, regularization and parallelization techniques to enhance predictive accuracy and computation efficiency, and as a result, it becomes a favored choice of most contestants in machine learning contests and also widely used in industrial predictive maintenance applications [7].

The objective function to be optimized by XGBoost is shown in Equation (3):

$$Obj = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{3}$$

where:

- Obj represents the overall objective function
- $l(y_i, \hat{y}_i)$ represents the loss function
- y_i represents the true label
- $\hat{y}_i$ represents the predicted label
- $\Omega(f_k)$ represents the regularization term
- n represents the number of training samples
- K represents the number of trees

The regularization term is defined by equation (4):

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda ||w||^2 \tag{4}$$

where:

- γ represents the complexity penalty parameter
- T represents the number of leaves
- λ represents the regularization coefficient
- w represents the leaf weight vector

Comparing the performance of the Random Forest and XGBoost allows us to gauge two strong ensemble methods for machine fault diagnosis. This study trained and tested both models using an identical pre-processing chain; this involved Remaining Useful Life calculation, generation of fault labels and re-sampling using the SMOTE technique to ensure class balance. Several classification measures, such as precision, recall, accuracy, F1-score and confusion matrix [8] analysis were utilized for their performance evaluation.

III. Methodology

Figure 1 represents the entire proposed work flow. Which includes data pre-processing, Remaining Useful Life (RUL) calculation, fault labeling, class balancing with SMOTE, machine learning training, performance testing and result visualization steps.

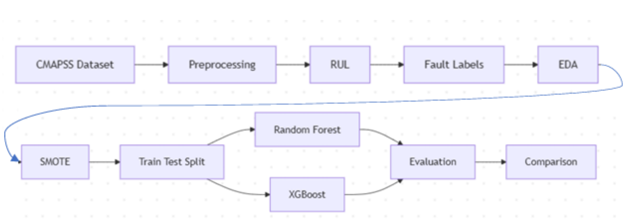

Fig. 1: Proposed methodology workflow

A. Data and Data Pre-processing

The dataset we used for this study is the NASA Commercial Modular Aero-Propulsion System Simulation (CMAPSS) dataset which was originally developed by Saxena et al [**?**]., which is popular used in predictive maintenance and Remaining Useful Life (RUL) prediction research. This dataset was created by NASA for turbofan engine degradation simulation and contains multivariate time-series sensor measurements collected under different operating conditions and fault scenarios.

before to feeding the CMAPSS training data sets (FD001, FD002, FD003 and FD004) into the suggested machine learning approaches, we merged these four data sets into one single comma-separated values file (CSV file) named *NASA_CMAPSS_Train_All.csv*. in order to have more diverse data by having more diverse operating conditions and different modes of fault occurring and to have more robust and well-generalized predictive models. The python codes applied for merging data sets and performing the preprocessing procedures can be found in the following **GitHub** address [10].

The extracted data contains the operational parameters, sensor measurements, engine cycle, and data set index from the NASA CMAPSS files. A row of the dataset represents an operation cycle of a turbofan engine.

The final dataset includes the following features:

- **unit_number**: represents the unit of the engine
- **time_cycles**: represents the operational cycle number of the engine
- **op_setting_1** and **op_setting_2** : represent the operational setting parameters affecting engine behavior
- **sensor_1** to **sensor_21**: represent sensor measurements collected during engine operation and used to monitor engine health and degradation behavior
- **dataset**: identifies the original CMAPSS subset (FD001, FD002, FD003, or FD004)
- **engine_id**: uniquely identifies each engine trajectory in the dataset

The measured features from the sensor provide information relevant for the status of the engine and the development of degradation in time. The features have been exploited as input variables in the training of machine learning models and fault identification.

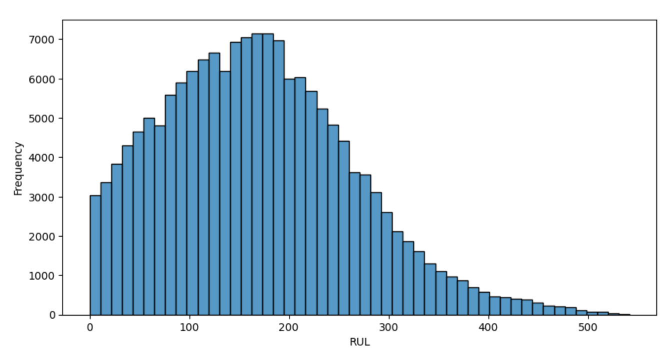

Fig. 2: Distribution of Remaining Useful Life (RUL)

Figure 2 shows the distribution of Remaining Useful Life values for all engine trajectories. RUL was calculated by subtracting the current cycle from the max cycle attained by the given engine (Equation (5)).

$$RUL = max\_cycle - time\_cycles \tag{5}$$

where:

- RUL represents the Remaining Useful Life
- max_cycle represents the final operational cycle before engine failure
- $time\_cycles$ represents the current operational cycle

This calculation shows how many working cycles are left for engine failure. Higher RUL means healthier engine. Small RUL means engine is close to failure. Figure 2 shows that the majority of the engine samples have medium RUL, with less number of observation under fully degraded.

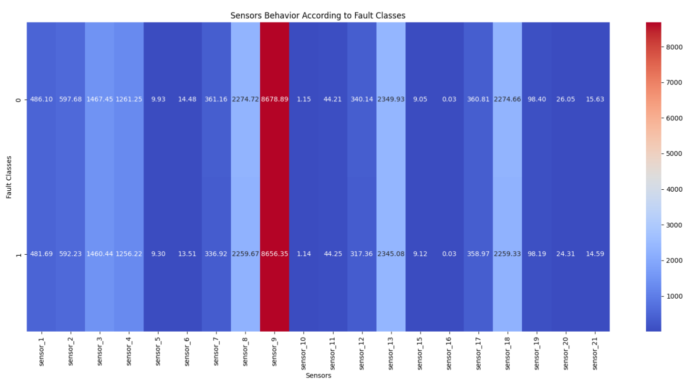

Fig. 3: Average sensor behavior according to fault classes

In figure 3 an average of sensors are shown for a good engine and a failed engine. We can see that for a number of sensors, there is a noticeable difference between good and failure behavior which indicates these are useful for diagnosis and prediction purposes. Some sensors change their behavior dramatically as we enter into failure regions while others do not vary at all

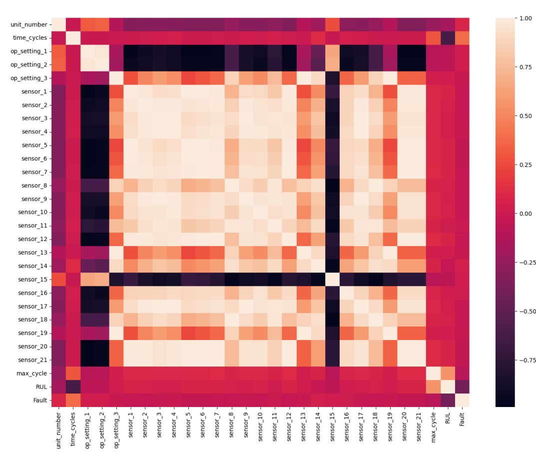

Fig. 4: Correlation matrix of operational settings and sensor measurements

The correlation matrix [11] of operation settings, sensor reading, Remaining Useful Life and fault labels is shown in Figure 4. It can be seen there is clear positive and negative correlation between a number of sensor readings, which indicates a dependency relationship between the measurements of the engine. Furthermore, it identifies which features are most correlated to engine degradation and fault states which will provide useful information when training machine learning models and classifying engine faults.

There were no distinct fault labels available in the initial NASA CMAPSS data set. For each engine trajectory, a Remaining Useful Life (RUL) was calculated as the difference between the highest cycle number and the current cycle number. With this value, we redefined the problem of predictive maintenance as a supervised binary classification task based on a fault labeling method using a threshold value. The samples with RUL 30 and below were defined as faulty and the remaining samples were defined as healthy operating condition.

The features that contain directly related future failure information was excluded before training process in order to assure realistical model training and avoid data leakage. Therefore, *RUL*, *max_cycle*, *engine_id*, and *dataset* are not included into training input feature set, and remaining operational settings and sensors' measurements were used as training input.

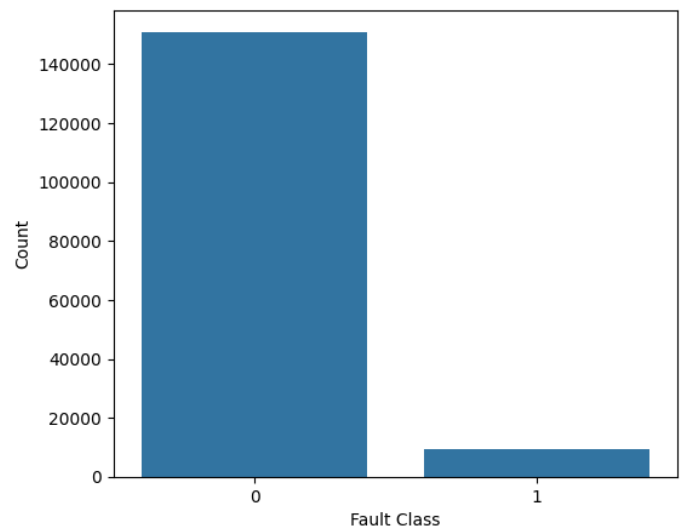

Fig. 5: distribution of the fault class before SMOTE

Figure 5 represents the distribution of faults in terms of fault classes prior to balancing. The original data was very imbalanced with about 94% of samples belonging to a healthy engine status (Class 0) while only 6% belonging to a faulty engine status (Class 1). This imbalance can cause problems for the performance of machine learning models since they would be likely to classify more often toward the majority class.

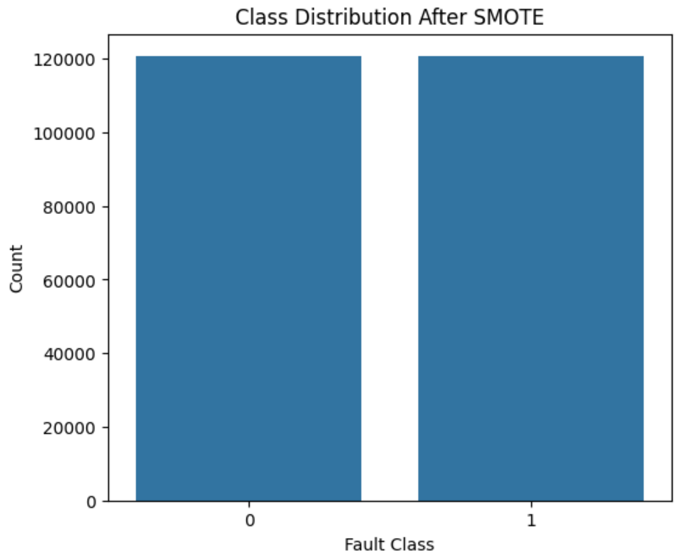

Fig. 6: distribution of the fault class after SMOTE

In order to resolve this problem, we used the Synthetic Minority Over-sampling Technique (SMOTE) [12] . SMOTE over-samples the minority class by creating synthetic data, thus balancing the data distribution. Figure 6 represents the distribution of the fault class after SMOTE was applied. The healthy and the faulty classes have become balanced, which allows the classification models to effectively predict the fault class and therefore increase the overall accuracy of prediction.

B. Model Selection

High predictive accuracy, resilience to noisy data and strong ability in capturing nonlinear relation between sensor values and engine degradation performance, respectively.

Random Forest was chosen for its excellent generalization performance and over-fitting resistance properties. The technique involves building an ensemble of decision trees based on a random sampling of the training instances and features and the output prediction is based on majority voting by each individual tree. Apart from its high accuracy, Random Forest can also analyze feature importance which can help to identify the critical sensors related to engine degradation.

XGBoost was chosen for its optimized gradient boosting framework and excellent classification accuracy in predictive maintenance context. The algorithm builds trees one by one and each tree is intended to correct the errors made by previous trees. It also embeds regularization and optimization which results in model accuracy and computational efficiency.

Two algorithms were trained on the balanced data set generated after using SMOTE. Hyperparameter tuning was conducted using RandomizedSearchCV for the two algorithms and determining optimal hyperparameter settings for the two models to boost the classification performance.

The optimized hyperparameters of the Random Forest model include the number of trees, maximum depth, the minimum number of samples per split, and minimum number of samples per leaf node, respectively. The optimized hyperparameters of XGBoost include the number of estimators, learning rate, and maximum tree depth.

The performance of these two models was measured by accuracy, precision, recall, F1-score, and confusion matrix to evaluate their ability in predicting healthy and faulty engine conditions.

IV. Experimentation and results

A. Experimentation

In this work, we performed various experiments in order to test the performance of Random Forest and XGBoost for intelligent machine fault classification using NASA CMAPSS data. Following pre-processing, computation of Remaining Useful Life, fault labeling, and balancing of classes using SMOTE, we then split the data into training and testing sets for the testing of the machine learning models.

In order to increase the predictive accuracy of the machine learning models, we ran hyper-parameter optimization using the RandomizedSearchCV [14]. Using this, we can automatically tune the hyper-parameters to get optimal parameters at increased computational cost compared to a grid search algorithm. We tuned the following hyperparameters for the Random Forest classifier: number of estimators, maximum depth of the tree and minimum samples split. The resulting optimal parameters are:

- *n_estimators = 200*
- *max_depth = 20*
- *min_samples_split = 2*

These optimized parameter sets resulted in an increased ability of the Random Forest classifier to capture the complex relationships that exist between the sensor measurements and engine fault states, while preserving a good ability to generalize.

In parallel, we carried out hyperparameter optimization of the XGBoost classifier using the randomized search with Cross-Validation technique. Optimized parameters among others included number of trees, learning rate, tree depth, and subsample ratio.

It turns out that optimal parameters for XGBoost were:

- *n_estimators = 300*
- *max_depth = 8*
- *learning_rate = 0.1*
- *subsample = 0.8*

The tuning of the XGBoost parameters allows us to have a better balance of the classification and a higher fault detection rate than Random Forest classifier. The use of the learning rate and subsampling helps us prevent overfitting and improves the generalization rate of the model. Finally, after finishing the tuning of the model, both models are trained on the balanced training data and their performances are evaluated on the classification accuracy, precision, recall, F1-score and on the confusion matrix.

After completing the optimization process, we trained both models using the balanced training dataset and evaluated their performance using accuracy, precision, recall, F1-score, and confusion matrix analysis.

We also made the implementation code and preprocessing scripts used in this work publicly available on **GitHub** [10] to ensure reproducibility and facilitate future research.

B. Random Forest Results

To assess its fault detection capabilities, we utilized the Random Forest classifier with classification metrics and the confusion matrix.

Based on our assessment, the Random Forest classifier obtained an overall accuracy of 95%. Specifically, the faulty engine class received precision of 52%, recall of 89%, and an F1-score of 66%. It appears the model was able to accurately identify many of the fault conditions but misclassified some healthy engine states as fault.

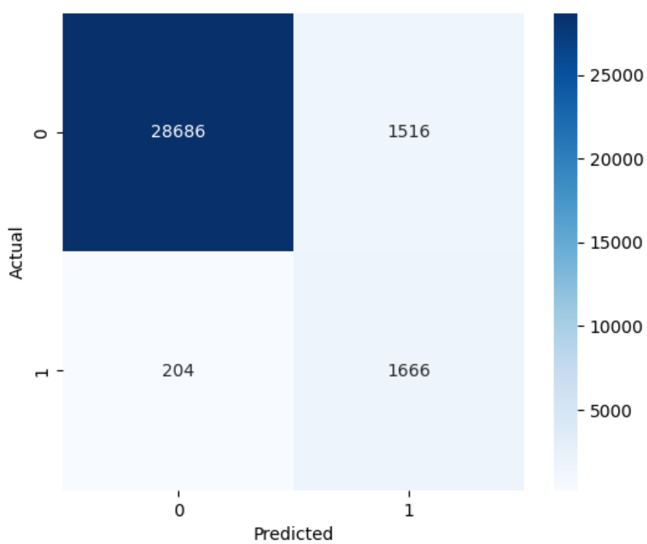

Fig. 7: Confusion matrix of the Random Forest model

Figure 7 represents the confusion matrix achieved from the Random Forest classifier. From the graph we can observe that the model accurately predicted 28,686 healthy engine states and 1,666 faulty engine states while mispredicting 1,516 healthy states as faults and 204 faulty states as healthy.

Overall the Random Forest classifier proved successful in detecting fault states and maintaining a good overall prediction value.

C. XGBoost Results

Similarly to Random Forest, we applied classification metrics and confusion matrix analysis to measure the performance of XGBoost.

Our analysis has shown that XGBoost classifier yielded an accuracy of 97%. It also performed much better with the precision of 70%, recall of 93%, and F1-score of 79% for the faulty engine class. This implies that XGBoost model provided balance between classification prediction and improved minority class prediction capability than the Random Forest model.

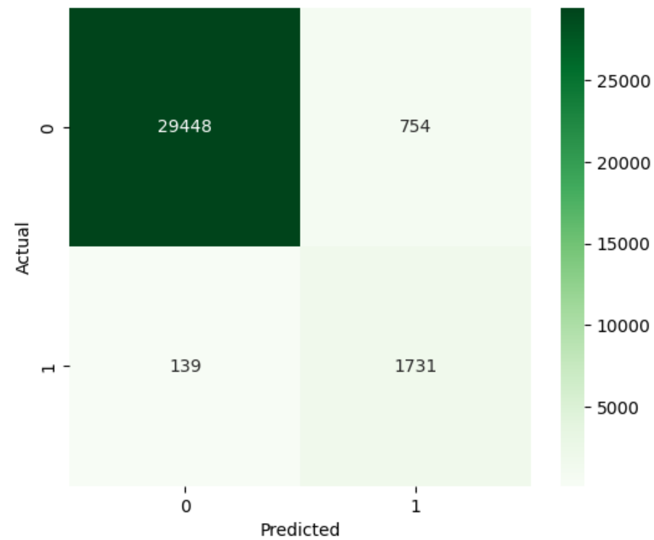

Fig. 8: Confusion matrix of the XGBoost model

Figure 8 shows the confusion matrix derived from the XGBoost classifier. We observed that it correctly predicted 29,448 healthy and 1,731 faulty states while mispredicting 754 healthy states as faults and 139 faulty states as healthy.

The confusion matrix produced by XGBoost resulted in fewer classification errors and overall higher accuracy compared to the Random Forest model. From the confusion matrix we confirm that XGBoost is more accurate in classifying between healthy and faulty engine state of NASA CMAPSS dataset.

D. Feature Importance analysis

We analyzed feature importance obtained by Random Forest and XGBoost models in order to find how much influence the sensor readings and operating parameters has on fault classification. Feature importance analysis is used to reveal variables contributing to predicting the machine fault and identifying engine deterioration using different measures [15]..

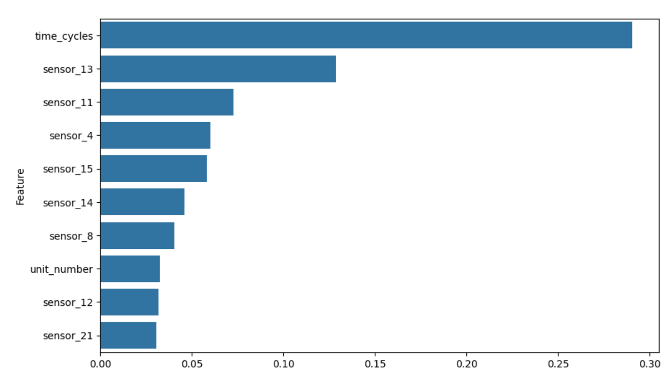

Fig. 9: Important features by Random Forest classifier

Figure 9 demonstrates the features that had the largest importance values obtained from the Random Forest classifier. The *time_cycles* has highest feature importance value; which means the operating cycle life of engine is of most interest. Several sensor readings such as *sensor_16*, *sensor_15*, *sensor_13* and *sensor_11* also exhibit high importance values indicating how these sensors contribute to fault classification. The analysis revealed the important sensors are more correlated to the engine deterioration and hence useful for both predictive maintenance and early fault detection applications.

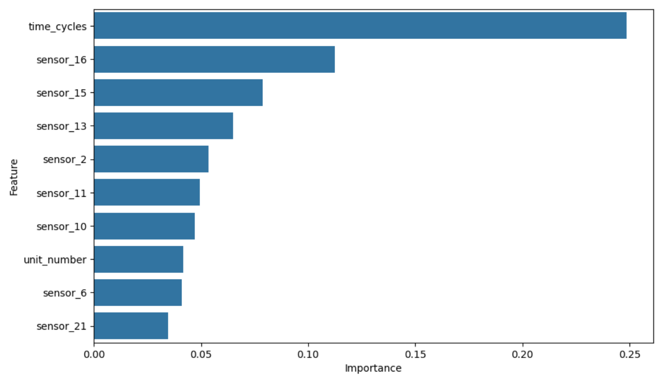

Fig. 10: Important features by XGBoost classifier

Figure 10 demonstrates feature importance analysis from the XGBoost classifier. Like in the Random Forest model, we discovered that *time_cycles* has highest feature importance value. We found that the sensors *sensor_13*, *sensor_11*, *sensor_4* and *sensor_15* also has high importance value in identifying engine deterioration. Compared to Random Forest, the importance value for key features in XGBoost is more centralized showing why the model has best prediction and fault detection accuracy.

V. Discussion

In this section, we discuss the experimental results obtained, analyze and compare the performance between Random Forest and XGBoost, evaluate the efficacy of the proposed method, and highlight the advantages and limitations of the developed fault classification system.

Figure 11 shows the comparison result between Random Forest and XGBoost in terms of accuracy, precision, recall and F1-score. Based on the experimental results obtained, we observed that both models produce favorable predictions when intelligent machine fault classification is carried out on the NASA CMAPSS dataset.

Random Forest produced good overall classification results and its recall ability on fault class is higher, meaning it detects fault classes properly. However, Random Forest's precision score is comparatively lower, which shows that it mistakenly predicted some normal engine conditions as fault conditions, though it remains highly robust and exhibits good generalization ability.

The overall classification performance between both models can be summarized as follows: XGBoost outperforms Random

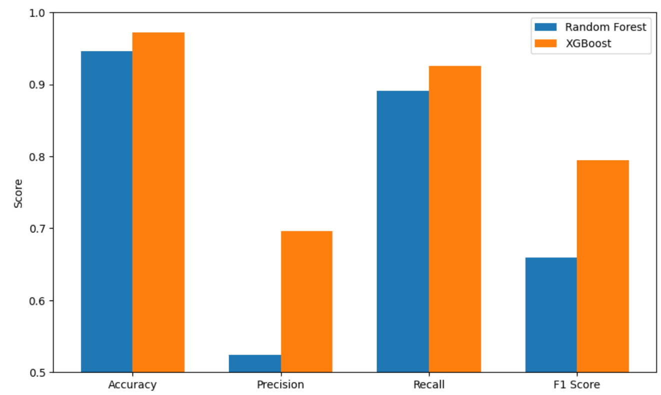

Fig. 11: Classification metrics comparison between Random Forest and XGBoost

Forest across all evaluated metrics. Precision, recall and F1-score for XGBoost are superior to the same parameters for Random Forest. This indicates that the gradient boosting method and the regularization utilized by XGBoost were the key features enabling higher fault detection efficiency and reduced misclassification.

Analysis based on the confusion matrix also showed that XGBoost achieved a lower number of false positive and false negative predictions compared to Random Forest. Therefore, it exhibits higher balance in classification performance.

One advantage of the proposed methodology is that we integrated all parts in an intelligent machine predictive maintenance cycle, including pre-processing, RUL calculation, fault label generation, oversampling for minority classes using SMOTE, hyper-parameter tuning and feature selection. In addition, using ensemble learning models increases robustness on noisy measurements from sensors and enhances classification performance.

Despite its successful applications, there are still limitations to this work. First, the fault labeling method uses a fixed threshold on RUL which does not necessarily represent a clear failure transition. Second, changing the problem to a binary fault classification task by utilizing RUL does not retain comprehensive information about the engine's health and degradation condition. Third, using SMOTE to synthesize minority samples can generate artificial patterns.

To conclude, we found that ensemble learning approaches, especially the XGBoost classifier, achieve good fault classification performance for intelligent machine predictive maintenance tasks using the NASA CMAPSS dataset.

VI. Conclusion and Future Work

In this study, we have designed and built an intelligent fault diagnosis system for the turbofan engine based on the NASA CMAPSS dataset of engine degradation. Our approach involves the following procedures; data preprocessing, Remaining Useful Life calculation, generation of fault labels, exploratory data analysis, class balancing using SMOTE, hyperparameter tuning, and machine learning models.

For prediction maintenance and fault classification of the engine degradation process, two ensemble learning models (Random Forest and XGBoost) were selected. From the obtained experimental results, both the machines predicted quite well at diagnosing healthy and faulty engine. In all performance metrics, it is found that the XGBoost model predicts best with high accuracy, precision, recall, and F1-score with minimum number of errors in the confusion matrix. The analysis of the features importance illustrated that the sensor readings of some engines and operation cycle had the maximum impact on diagnosing and predicting of the engine degradation and fault condition. In addition, it is concluded that SMOTE balancing algorithm is well at diagnosing and predicting minority faults and reduces the prediction bias for majority fault type.

From the result, it can be said that ensemble machine learning algorithms are efficient in applying to the intelligent prediction maintenance application and machine fault diagnosis based on multi-sensor data.

However, there are several limitations for this work. First, the fault label was generated based on a specified RUL threshold; thus it is possible that some engines are considered as faults, while others remain healthy, despite that they both have similar degradation patterns. Second, reformulating the predictive maintenance into a classification problem is a coarse estimation and loses some of the detail of the degradation information.

More advanced methods like Long Short-Term Memory (LSTM) or Transformer neural networks or hybrid deep learning methods would be examined in future studies for Remaining Useful Life prediction and multi-stage diagnosis. Also, research may focus on multiclass fault prediction, on-line predictive maintenance systems or interpretable AI models.

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